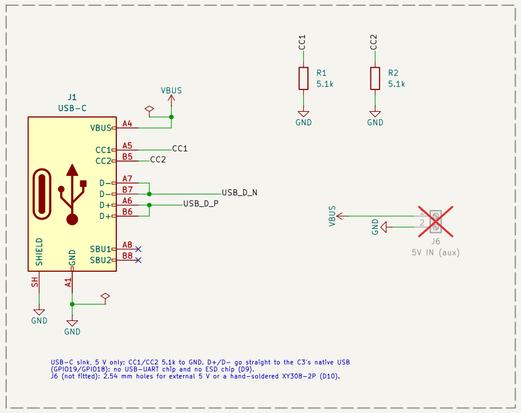
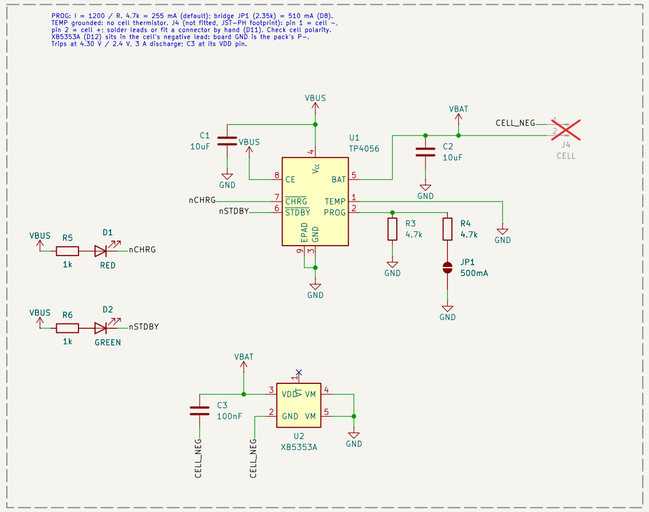
USB-C: 5 V IN AND PROGRAMMING

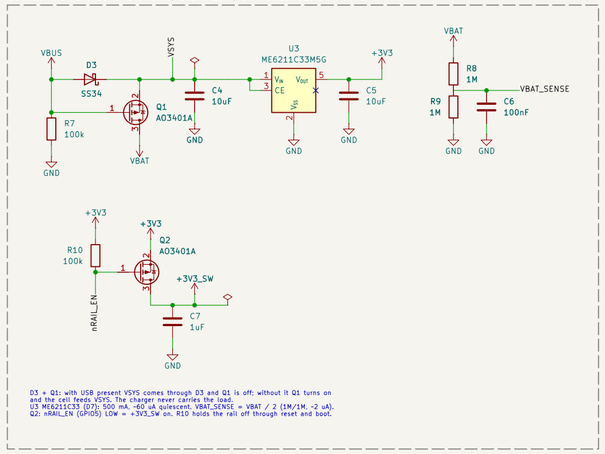


USB-C sink. 5 V only: CC1/CC2 5.1k to GND. D+/D- go straight to the C3's native USB (GPIO19/GPIO18); no USB-UART chip and no ESD chip (D9).
J6 (not fitted): 2.54 mm holes for external 5 V or a hand-soldered XY308-2P (D10).

LI-ION CHARGER & CELL PROTECTION

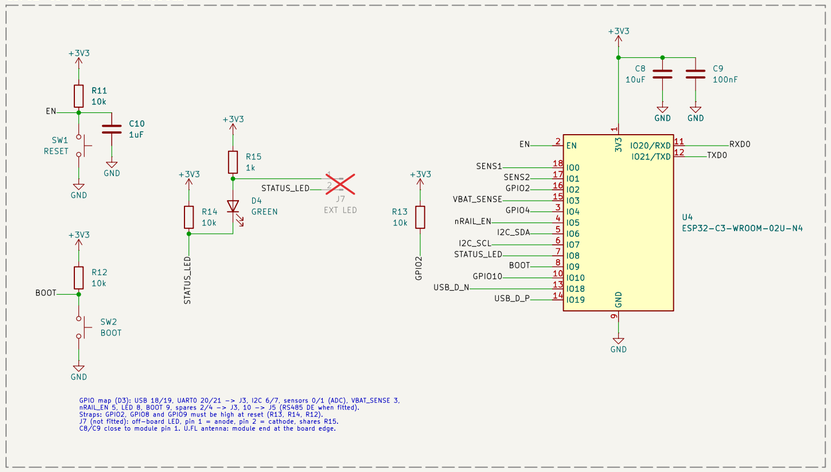


SYSTEM POWER: LOAD SHARING, 3.3 V LDO, SWITCHED SENSOR RAIL



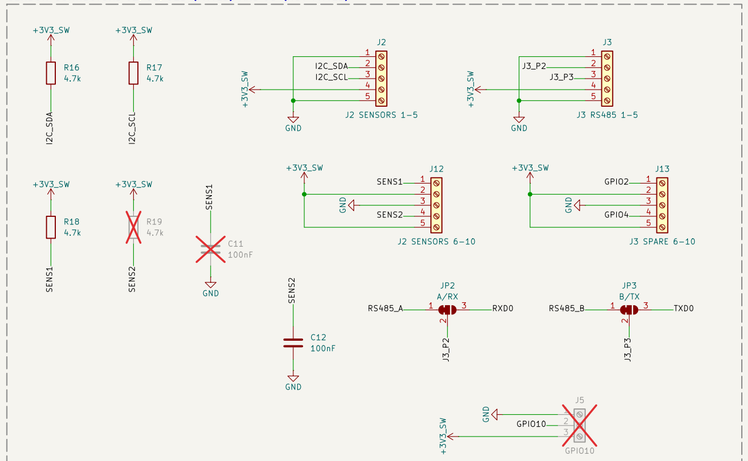
D3 + Q1: with USB present VSYS comes through D3 and Q1 is off; without it Q1 turns on and the cell feeds VSYS. The charger never carries the load.
U3 ME6211C33 (D7): 500 mA, ~60 uA quiescent. VBAT_SENSE = VBAT / 2 (1M/1M, ~2 uA).
Q2: nRAIL_EN (GPIO5) LOW = +3V3_SW on. R10 holds the rail off through reset and boot.

ESP32-C3-WROOM-02U MODULE



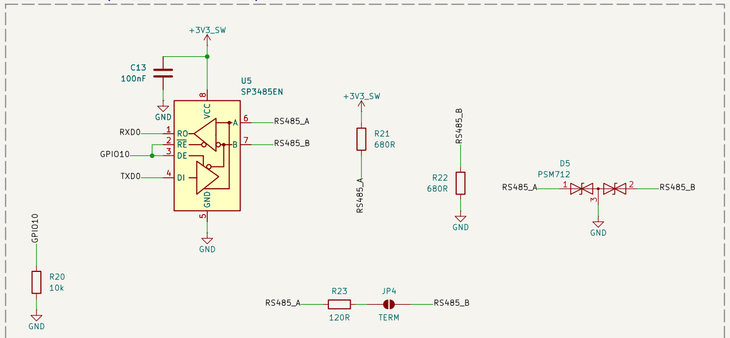
GPIO map (D3): USB 18/19, UART0 20/21 -> J3, I2C 6/7, sensors 0/1 (ADC), VBAT_SENSE 3, nRAIL_EN 5, LED 8, BOOT 9, spares 2/4 -> J3, 10 -> J5 (RS485 DE when fitted).
Straps: GPIO2, GPIO8 and GPIO9 must be high at reset (R13, R14, R12).
J7 (not fitted): off-board LED, pin 1 = anode, pin 2 = cathode, shares R15.
C8/C9 close to module pin 1; U.FL antenna: module end at the board edge.

SENSOR CONNECTORS J2, J3, J13 (FITTED) AND J5 (NOT FITTED)



J2 (main) and J3 share one layout, pins 1–10 (D4, D5): J2: J12, J3 and J13 all fitted:
1 GND 2 SDA/A 3 SCL/B 4 3V3_SW 5 GND 6 S1/GPIO2 7 3V3_SW 8 GND 9 S2/GPIO4 10 3V3_SW
Each is two XY308-2.54-5P blocks side by side (no 10-way part at JLCPCB): J2=J12 . J3=J13.
JP2/JP3: J3 pins 2–3 = RS485 A/B as delivered; cut 1–2 and bridge 2–3 for UART0 RX/TX (U5 must then be left off) (D29).
J5 (not fitted): GND / GPIO10 / 3V3_SW (D14). R18 (SENS1 pull-up) and C12 (SENS2 filter) fitted; R19, C11 not fitted.
Firmware: switch +3V3_SW on (GPIO5 low) before any I2C or sensor access.

RS485 / MODBUS (FITTED; OMIT TO SAVE COST)



Fitted by default: J3 pin 2 = A, pin 3 = B via JP2/JP3 (D29). A board without Modbus can leave out U5, C13, R20–R23, D5 and J3 (see COST-ESTIMATE.txt). GPIO10 = DE/RE (high = transmit, R20 = receive).
JP4 bridged = 120R termination. RS485 sensors need their own 5–24 V supply (external).

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Natural Innovation

Sheet: /
File: frugal_iot_v02.kicad_sch

Title: Frugal-IoT V0.2

Size: A2	Date: 2026-10-22	**Rev: 0.2**
KiCad E.D.A. 10.0.4		Id: 1/1